

Vijay Singh

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EDUCATION

Indian Institute of Science, Bengaluru <i>M.Tech in Transportation Systems Engineering</i>	<i>Aug 2024 – Jul 2026</i> CGPA: 7.9 / 10
Punjab Technical University <i>B.Tech in Civil Engineering</i>	<i>Aug 2016 – Jul 2020</i> CGPA: 8.8 / 10

WORK EXPERIENCE

IIT Patna: Junior Research Fellow (JRF) — EDAS Project • Working on reinforcement learning (RL)-based approaches for optimization and decision-making under the Eco Driving Assisted Systems (EDAS) project.	<i>July 2026 – Present</i>
Chegg: Subject Matter Expert - Freelance • Prepared academic solutions for Civil Engineering and Psychology problems.	<i>Jul 2023 – Aug 2023</i>
CADD Centre: Technical Software Trainer • Trained students in CAD-based project work and conducted visiting classes at colleges.	<i>Sep 2022 – Mar 2023</i>
Secretary to NCC Associate NCC Officer • Assisted drill, parade, and training sessions, supported training camps with Indian Army and maintained records & documentation.	<i>Jul 2020 – Jul 2022</i>

INTERNSHIPS

TIA Project, IISc: Bagchi-Parthasarathy Hospital • ANPR camera location planning, hospital site selection, data collection survey, draft bill preparation and VISSIM/SUMO simulation.	<i>Jun 2025 – July 2025</i>
SJVN: Civil Engineering Intern • BOQ and DPR review, quantity reconciliation, site measurement verification, and work progress reporting.	<i>Apr 2023 – Jun 2023</i>
Deepak Builders: Road and Bridge Construction Intern • Pathankot-Chakki Bank road and bridge construction stretch: site survey, material testing, mix checking, and field measurements.	<i>Apr 2020 – Jun 2020</i>
CADD Centre, Pathankot: • CAD software learning and project preparation using AutoCAD, 3ds Max, Revit and STAAD Pro.	<i>Jan 2020 – Mar 2020</i>

SKILLS

Python, R | ML, Choice Modeling | Gurobi, NetworkX, Google OR-Tools | VISSIM, SUMO | QGIS, Power BI | Pandas, NumPy, Matplotlib, Excel, LaTeX

COURSES

Optimization Methods | Logistics & Freight Modeling | Traffic Network Equilibrium | Choice Modeling | Transportation Systems Modelling | Machine Learning for Cyber-Physical & Mobility Systems | Introduction to Multi-Modal Mobility Systems

M.TECH THESIS

Thesis Title: Estimation of Origin-Destination Matrices Using Traffic Counts [\[Report\]](#) [\[Presentation\]](#)

Importance: Estimated OD matrices enable dynamic tracking of travel demand, early bottleneck identification, real-time signal timing and route optimization — building a continuously updating pipeline from junction-level camera data and replacing costly manual traffic surveys.

PROJECTS

Branch-and-Cut Algorithm for GTSP

Python, Gurobi, Matplotlib

- Formulated the Symmetric Generalized Travelling Salesman Problem (GTSP) ; implemented and compared baseline MILP, construction heuristic with 2-opt and Branch-and-Cut using subtour elimination constraints.

Branch-and-Bound (B&B) for GAP and UFLP

Python, Gurobi

- Formulated the Generalized Assignment Problem (GAP) and Uncapacitated Facility Location Problem (UFLP)

Branch-and-Cut (B&C) for Symmetric TSP

Python, Gurobi, NetworkX, OR-Tools

- Implemented a B&C framework for the Symmetric Travelling Salesman Problem (STSP) using connectivity cuts and 2-connected cuts ; benchmarked performance against Google OR-Tools.

Contributor Verification using Machine Learning

Python, ML

- Developed a binary classification model to verify article contributors ; created count-based features and trained an XGBoost model

Last-Mile Mobility Data Analysis and Visualization

Python, Excel, Power BI, QGIS

- Analyzed metro last-mile survey data for Bengaluru, Delhi and Nagpur by cleaning records ; developed Power BI dashboards and QGIS maps to visualize mode share, travel time, travel cost, distance and station-level accessibility patterns.

Discrete Choice Modeling: Swissmetro Mode Choice

MNL, Nested Logit, MMNL

- Estimated MNL, Nested Logit, and Mixed Multinomial Logit (MMNL) models in R Studio on the Swissmetro stated-preference dataset to model mode choice among Train, Car and high-speed Swissmetro.

ACHIEVEMENTS & ACTIVITIES

- Attended the Large-Scale Optimization (LSO) Summer School at IIT Delhi in Jan 2026
- Attended the World Symposium on Sustainable Transport & Livability (WSSTL-2025) at IISc in Jun 2025
- Participated in Spectrum (Inter-Departmental Sports Meet at IISc) in 2025 & 2026, winning multiple medals
- Secured AIR 406 in the GATE 2024 Civil Engineering examination
- Secured University Rank 1 at Punjab Technical University for the academic session 2017
- Awarded the Best Manager Award by the Department of Management, CGC University, in 2020
- Awarded the Best Company Commander by NCC 23 Punjab Battalion in 2018